

Assessment Test: Data Analyst

Duration: 10 hrs

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Business Context

Assume you are working as a **Data Analyst at a B2B technology company**.

Management wants to become more data-driven and has asked you to identify **business opportunities, performance gaps, trends, or operational improvements** using publicly available data.

You have not been given a predefined dataset.

Your task is to independently identify a suitable dataset, define a meaningful business problem, analyze the data, and present your findings to management.

You may choose a dataset related to areas such as:

- Sales
- Retail
- E-commerce
- Customer behavior
- Marketing
- Restaurants / Food & Beverage
- Transportation
- Hospitality
- Finance
- Operations
- Healthcare
- Technology
- Any other relevant business domain

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Question 1

Find **ONE large, publicly available dataset** from a credible source and transfer it to a Google Sheet (**Worksheet Name: Data**).

The dataset should ideally contain:

- **50,000+ records**, OR
- Multiple related datasets/files, OR
- Sufficient complexity to require meaningful data processing and analysis.

Submit (Worksheet Name: Q1)

1. Dataset name
2. Source and URL
3. Number of rows and columns
4. Brief description of the dataset
5. Why you selected this dataset
6. What business opportunities or problems you believe the dataset could help investigate

Question 2

Based on your selected dataset, assume you are presenting the analysis to **business management**.

Define (Worksheet Name: Q2):

- A. What business problem or opportunity are you trying to understand?
- B. Define **3–5 questions** that your analysis will answer.
- C. Define at least **2 hypotheses** that you want to test.

Question 3

Process the raw dataset (**Worksheet Name: Processed Data**) using **ONE** of the following approaches:

Option A – Python

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook
- Google Colab

Option B – Google Sheets + Apps Script

- Google Sheets
- Apps Script
- Built-in spreadsheet functions



Your processing should include relevant activities such as:

- Handling missing values
- Removing duplicates
- Correcting data types
- Standardizing data
- Creating calculated fields
- Joining/merging data where applicable
- Categorizing data
- Aggregation
- Identifying unusual/outlier records
- Preparing an analysis-ready dataset

Explain (Worksheet Name: Q3):

- What did you change in the raw data and WHY?
- For at least **3 major data-cleaning decisions**, explain:

Change	Why was it necessary?	What would happen if you didn't do it?
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Question 4

Perform exploratory and descriptive analysis on your processed dataset. Identify **at least 5 meaningful business insights (Worksheet Name: Q4)**. For each insight, use the following structure:

Insight	Evidence	Why Did You Select This Insight?	Business Impact	Recommendation
What did you discover?	What data/calculation/analysis supports your conclusion?	Why is it relevant to management?	What could be the potential impact or opportunity?	What action should management consider?

Question 5

During your analysis, identify **at least ONE surprising, unusual, or unexpected result (Worksheet Name: Q5)**.

For example:

- An apparently weak segment performs better than expected.
- A high-sales segment has poor profitability.
- A smaller customer group generates disproportionately high value.
- A trend changes unexpectedly over time.
- Two variables behave differently from what you initially expected.

Explain:

1. What did you initially expect?
2. What did the data actually show?



3. Why do you think this happened?
4. What additional analysis did you perform?
5. What conclusion can you reasonably draw?

If the data does not allow you to determine the exact reason, **say so**.

Question 6

Identify at least **3 data-quality issues, limitations, or analytical risks** in your dataset (**Worksheet Name: Q6**).

For each:

- Identify the issue
- Explain how it could affect your analysis
- Explain how you handled or accounted for it

Also identify:

- A. Two limitations of your analysis
- B. One conclusion that CANNOT safely be made from your dataset

Question 7

Based on your analysis, provide **3 actionable recommendations** for management (**Worksheet Name: Q7**).

For each recommendation, explain:

- What should be done?
- Which insight supports the recommendation?
- Who would need to act?
- What potential business outcome could it create?
- What would you measure to determine whether the recommendation worked?

Prioritize recommendations based on **business impact and feasibility**.

Question 8

Create an interactive **Google Looker Studio** dashboard using your processed/analysis-ready data. Your dashboard should be designed for **business management**, not for another data analyst.

It should ideally include:

- Executive KPI summary
- Relevant trends
- Segment/category analysis
- Comparisons
- Filters
- Visualizations supporting your key findings



Submit (Worksheet Name: Q8):

1. Screenshot(s) of your dashboard
2. Shareable Looker Studio link, where possible
3. A short explanation of the purpose of your major visualizations

Important:

- **Clarity > Number of Charts**
- Do not create charts simply to make the dashboard look impressive.
- Every important visualization should help answer a business question.

Question 8

Prepare a **5–7 slide presentation** for senior/business management. Your presentation should follow approximately this structure:

- **Slide 1 – Business Problem:** What problem/opportunity did you investigate?
- **Slide 2 – Data & Methodology:** What data did you use and how did you analyze it?
- **Slide 3 – Key Findings:** What are the most important things management should know?
- **Slide 4 – Deep-Dive Insight:** Explain your most important finding with supporting evidence.
- **Slide 5 – Recommendations:** What should management do?
- **Slide 6 – Expected Business Impact:** What could improve and how would you measure success?
- **Slide 7 – Limitations & Next Steps:** What should management be cautious about and what should be analyzed next?

You should be able to explain your findings to a **non-technical management audience** without relying on technical terminology.

Question 8

AI tools are **allowed and encouraged** during this assessment.

You may use AI for:

- Understanding unfamiliar datasets
- Writing or improving code
- Debugging
- Brainstorming analytical approaches
- Research
- Documentation
- Visualization ideas

However, you are responsible for everything you submit. Include a short section explaining:

How I Used AI? (Worksheet Name: Q10)

Mention:

1. Which AI tools you used



2. What you used them for
3. One example where AI helped you
4. One example where you had to **verify, correct, or modify** an AI-generated result

Final Submission

Create **ONE Google Drive folder** containing the following:

1. **One Google Sheet** includes all worksheets created
2. **Code:** Python notebook/script OR Google Apps Script.
3. **Analysis:** Supporting calculations, analysis files, or notebooks. (if additional)
4. **Presentation:** 5–7 slide management presentation.
5. **README / Methodology**

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